Faculty Of Computer and Data Science

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Humanoid Football-Kicking Robot

Embedded Systems Course Project

Prepared for

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By

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Report Agenda

Chapter 1: Introduction

1.1 Historical Background

Humanoid robots are robots designed to imitate human body movement and behavior. In recent years, robotics has become one of the most important fields in engineering and artificial intelligence due to its applications in automation, healthcare, manufacturing, and entertainment.
Football-playing robots are considered one of the most challenging robotics applications because they require balance, walking control, object detection, and real-time decision making. Competitions such as RoboCup encouraged the development of autonomous humanoid robots capable of walking, tracking a football, and performing kicking actions.
With the development of embedded systems, computer vision, and machine learning algorithms, it became possible to create low-cost humanoid robots using microcontrollers, servo motors, and AI-based vision systems.

1.2 Problem Statement

Designing a humanoid robot capable of walking and kicking a football is considered a difficult engineering problem because the robot must maintain stability while moving and accurately detect the football using computer vision.
The system also requires accurate synchronization between the mechanical structure, embedded electronics, and software algorithms. Multiple servo motors must be controlled precisely to achieve smooth walking and kicking motions.
This project aims to solve these challenges by developing a wireless humanoid football-kicking robot capable of:

- Detecting a football using computer vision.
- Calculating movement angles using inverse kinematics.
- Walking toward the football.
- Performing a kicking motion.
- Operating wirelessly after uploading the code.

1.3 Aims and Objectives of the Project

The main objectives of this project are:

- Design a low-cost humanoid robot.
- Implement stable walking and kicking motions.
- Use computer vision for football detection.
- Apply inverse kinematics calculations.
- Develop a wireless embedded robotic system.
- Integrate hardware and software into a complete robotics platform.

1.4 Report Agenda

This report is divided into six chapters.
Chapter 1 introduces the project background, problem statement, and objectives.
Chapter 2 presents a literature review discussing previous humanoid robot systems and their limitations.
Chapter 3 explains the proposed system architecture and improvements over existing approaches.
Chapter 4 discusses the implementation details including mechanical design, electronics, software, and state machines.
Chapter 5 presents system results, kinematics accuracy, and vision performance measurements.
Chapter 6 concludes the project

Chapter 2: Literature Review

2.1 Existing Humanoid Robot Designs

Humanoid robots have been widely studied in robotics research due to their ability to imitate human motion. Most humanoid systems use multiple servo motors and articulated joints to achieve walking and balancing movements. Many research projects focus on football-playing humanoid robots because they combine locomotion, computer vision, and intelligent control.

2.2 Mechanical Design Approaches

Previous humanoid robots commonly use multiple degrees of freedom to improve flexibility and movement accuracy. Some systems use lightweight materials such as aluminum or 3D printed plastic to reduce robot weight and improve motion efficiency.
Walking mechanisms in humanoid robots are generally based on alternating leg movement while maintaining center-of-mass balance.

2.3 Electronics and Embedded Systems

Embedded systems are essential for humanoid robot control. Microcontrollers such as Arduino, Raspberry Pi, and ESP32 are commonly used in robotics applications.
Servo drivers such as the PCA9685 are frequently used to generate accurate PWM signals for controlling multiple servo motors.
Wireless communication and battery-powered operation are also important features in modern humanoid robots.

2.4 Computer Vision and Object Detection

Computer vision systems allow humanoid robots to interact with their environment.
Traditional object detection methods used color filtering and image segmentation. However, modern systems use deep learning models such as YOLO because they provide faster and more accurate object detection.
YOLO is suitable for real-time robotics applications because it can process images quickly while maintaining high detection accuracy.

2.5 Problems and Limitations of Previous Systems

Several limitations exist in previous humanoid robot systems:

- High system cost.
- Complex mechanical structures.
- Stability problems during walking.
- Servo motor overload.
- False object detection.
- Motion blur caused by camera movement.
- Limited processing speed.

Some systems also require wired operation or expensive hardware components.
The proposed project attempts to reduce system cost while maintaining acceptable walking and detection performance.

Chapter 3: Proposed System

3.1 System Overview

The proposed system is a wireless humanoid football-kicking robot capable of detecting a football using computer vision and moving toward it automatically.
The robot integrates:

- Mechanical structure.
- Embedded electronics.
- Computer vision.
- Inverse kinematics.
- Servo motor control.

The system performs football detection, walking, and kicking operations autonomously.

3.2 Improvements Over Existing Humanoid Approaches

The proposed system introduces several improvements:

- Low-cost implementation.
- Lightweight 3D printed structure.
- Wireless operation.
- Simplified electronics architecture.
- Real-time YOLO-based football detection.
- Accurate PWM generation using PCA9685.

The design reduces system complexity while maintaining functionality.

3.3 Hardware Architecture

The hardware system consists of:

- ESP32 microcontroller.
- PCA9685 servo driver.
- Six MG995 servo motors.
- Webcam.
- Bearings
- 12V Li-ion battery.
- LM2596 buck converter.
- Mechanical leg structure.

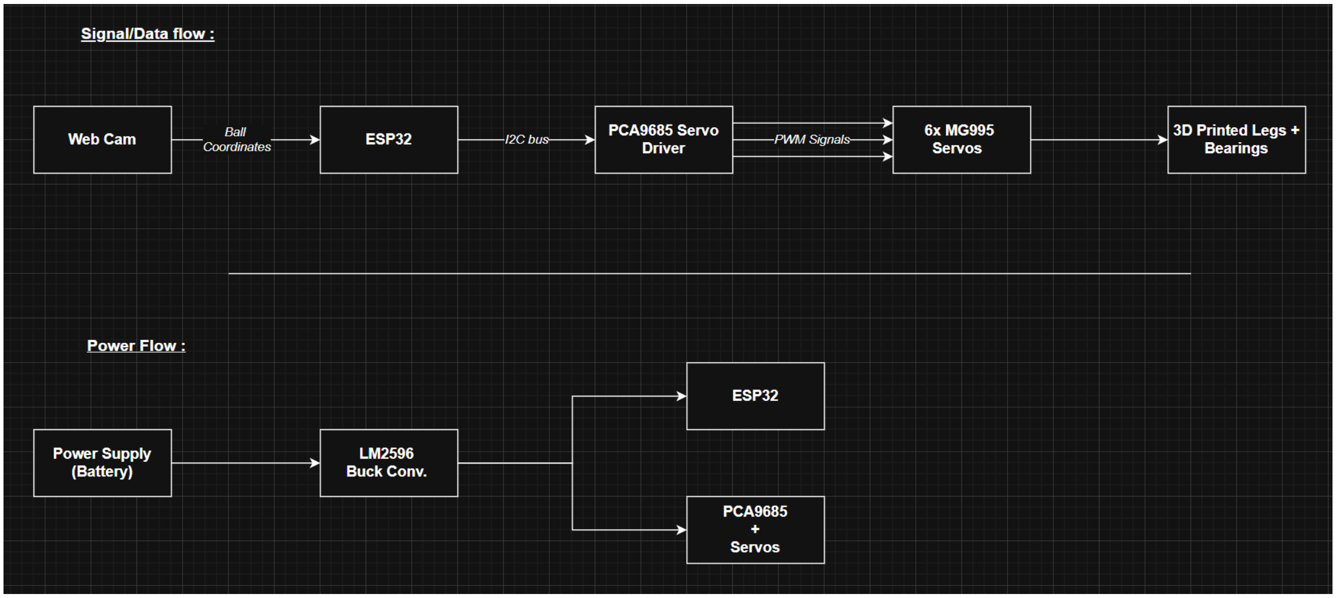
The ESP32 communicates with the PCA9685 through the I2C protocol.

3.4 Software Architecture

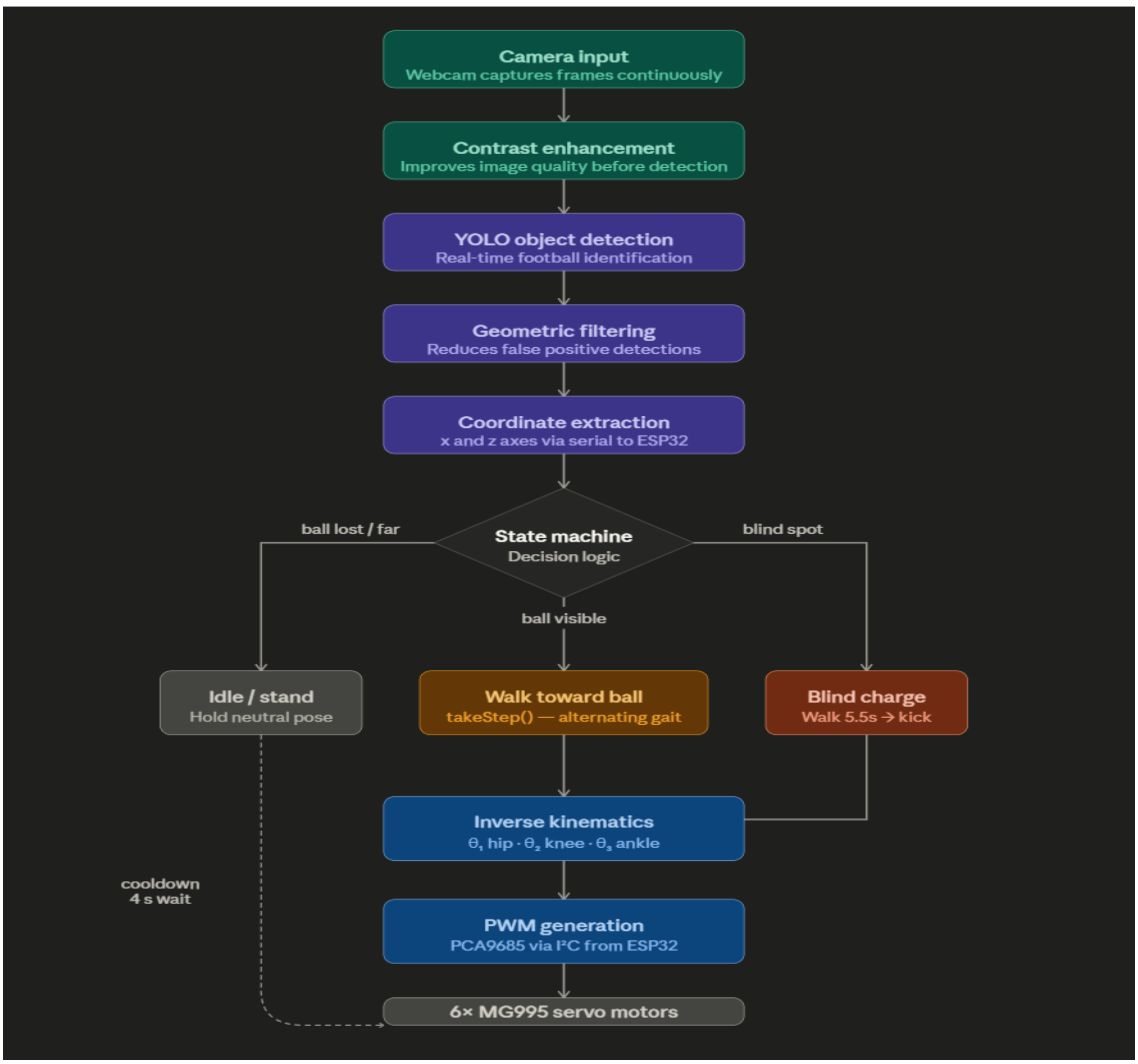
The software system consists of:

1. Camera image acquisition.
2. YOLO object detection.
3. Coordinate extraction.
4. Inverse kinematics calculations.
5. PWM signal generation.
6. Servo movement execution.

3.5 Hardware Block Diagram



3.6 Software Block Diagram



Chapter 4: Implementation

4.1 Mechanical Design

The robot structure was designed using MATLAB and manufactured using 3D printing technology.
Plastic material was selected because it is lightweight and suitable for low-cost robotics applications.
Each robot leg contains three degrees of freedom:

- Hip joint
- Knee joint
- Ankle joint

The robot uses six MG995 servo motors distributed equally between both legs.
The dimensions of the leg segments are:

- L1 = 5 cm
- L2 = 5.7 cm

Bearings were added beside the joints to reduce friction and improve movement smoothness.

4.2 Electronics Implementation

The electronics system controls the robot movement and power distribution.
The robot uses:

- ESP32 microcontroller.
- PCA9685 servo driver.
- 12V Li-ion battery.
- LM2596 buck converter.

The PCA9685 was selected because the ESP32 has limited stable PWM outputs for multiple servos.
The PCA9685 generates accurate PWM signals independently and prevents ESP32 overload.

4.3 Software Implementation

The project software was developed using Arduino IDE and Python.
The following Python libraries were used:

- import cv2
- import numpy as np
- import time
- import json
- import os
- import serial
- from ultralytics import YOLO

The camera captures images continuously and sends them to the YOLO object detection model.
The detected football coordinates are processed using inverse kinematics calculations to determine the required servo angles.

4.4 State Machines

The robot behavior can be represented using state machines.
The robot states include:

1. Idle State

2. Ball Detection State
3. Walking State
4. Position Adjustment State
5. Kicking State
6. Stop State

The system transitions between these states automatically depending on football position and robot movement.

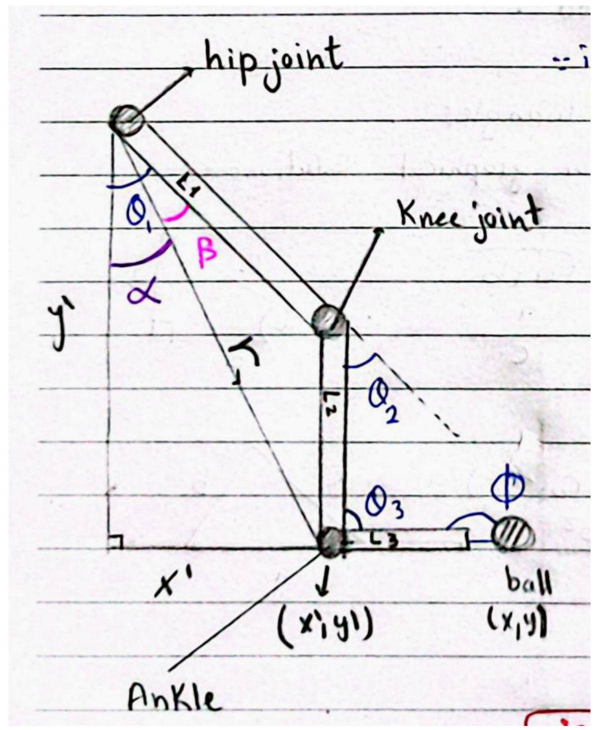
4.5 Inverse Kinematics Implementation

Inverse kinematics calculations are used to determine the required servo angles based on the football coordinates. The calculations use:

- Law of Cosines
- Inverse Tangent equations

The system calculates:

- Hip angle = theta1
- Knee angle = theta2
- Ankle angle = theta3



$$\alpha = \tan^{-1}\left(\frac{x'}{y'}\right)$$

$$\beta = \cos^{-1}\left(\frac{L_1^2 + r^2 - L_2^2}{2L_1 r}\right)$$

$$\theta_1 = \alpha + \beta$$

$$\theta_2 = 180^\circ - x^\circ = 180 - \cos^{-1}\left(\frac{L_1^2 + L_2^2 - r_2^2}{2L_1L_2}\right)$$

$$\begin{aligned}
\theta_3 &= 180 - a_1 - a_2 \\
&= 180 - (90 - \alpha) - a_2 \\
&= 180 - 90 + \alpha - a_2 \\
&= 90 + \alpha - a_2 \\
&= 90 + \tan^{-1}\left(\frac{X}{y}\right) - \cos^{-1}\left(\frac{L_2^2 + r^2 - L_1^2}{2L_2 r}\right)
\end{aligned}$$

4.6 Servo Control Implementation

Servo motors are controlled using PWM (Pulse Width Modulation) signals generated by the PCA9685 servo driver. The ESP32 sends angle commands to the PCA9685 through the I2C communication protocol. The PCA9685 converts these commands into accurate PWM signals for the six MG995 servo motors.
The servo angles are constrained between 0° and 180° to avoid mechanical damage and unstable movement.
The following function was used to generate PWM signals:

```
void writeServo(int channel, int angle) {
  angle = constrain(angle, 0, 180);
  pwm.setPWM(channel, 0, map(angle, 0, 180, SERVOMIN, SERVOMAX));
}
```

The generated PWM signals control:

- Hip joint movement
- Knee joint movement
- Ankle joint movement

Servo offsets were also applied to compensate for mechanical differences between the left and right legs.

4.7 Vision System Implementation

The vision system is responsible for detecting the football and extracting its coordinates.
A webcam continuously captures image frames which are processed using the YOLO object detection algorithm.
Before detection, contrast enhancement is applied to improve image quality and increase detection reliability.
The vision processing pipeline consists of:

1. Image Acquisition
2. Contrast Enhancement
3. YOLO Detection
4. Object Tracking
5. Geometric Filtering
6. Coordinate Extraction

The geometric filtering stage reduces false positive detections by validating the detected football shape and position.
The extracted coordinates are represented using the x and z axes and transmitted to the ESP32 controller through serial communication.
The vision system operates at approximately:

- 15 FPS
- 66 ms coordinate update interval

Despite achieving successful detection, some limitations were observed during testing. Rapid robot movement caused camera motion blur, making object detection more difficult under certain conditions.

4.8 Bill of Materials (BoM)

Component	Quantity	Cost (EGP)
MG995 Servo Motor	6	1020
PCA9685 Servo Driver	1	210
ESP32 Dev Module	1	280

Component	Quantity	Cost (EGP)
LM2596 Buck Converter	1	70
Breadboard	1	25
3D Printed Parts	1 Set	450
Bearings	6	250
Webcam	1	Already Available
Jumper Wires	Multiple	Already Available

Total Cost:
2305 EGP (excluding reused components)

Chapter 5: Results

5.1 Inverse Kinematics Accuracy

Inverse kinematics calculations were successfully used to determine the required hip, knee, and ankle angles for walking and kicking movements.
The calculated angles produced stable movement for most walking cycles. However, small deviations between theoretical and actual movement were observed due to servo limitations and mechanical instability.
The inverse kinematics implementation successfully converted the detected football coordinates into servo motor movements.

5.2 Vision Latency Measurement

The vision system achieved real-time football detection using YOLO object detection.
The system processed images at approximately:

- 15 frames per second
- 66 milliseconds coordinate refresh rate

The latency was considered acceptable for humanoid robot movement and football tracking applications.
Some additional delay occurred because of:

- Serial communication
- Image preprocessing
- Robot movement vibration

5.3 Detection Accuracy

The YOLO-based detection system successfully identified the football under normal lighting conditions.
The geometric filtering stage reduced false positive detections and improved tracking reliability.
Detection accuracy decreased during:

- Fast robot movement
- Sudden lighting changes
- Motion blur conditions

The system performed best under stable indoor lighting conditions with moderate robot speed.

5.4 Walking Performance

The robot successfully performed alternating leg walking motions and moved toward the detected football.
The walking gait was based on moving one leg while the other leg remained fixed to maintain stability.
The robot was able to:

- Walk forward
- Detect the football
- Adjust movement direction
- Perform a kicking action

However, some walking instability and drifting were observed because of mechanical balance limitations and servo inaccuracies.

5.5 Challenges During Testing

Several challenges were encountered during system implementation and testing.
Mechanical Challenges:

- Robot instability
- Slipping during walking
- Joint vibration

Vision Challenges:

- False football detections
- Motion blur caused by camera movement
- Sensitivity to lighting conditions

Control Challenges:

- Walking drift
- Servo synchronization errors
- Difficulty maintaining stable balance during kicking

Despite these challenges, the robot successfully demonstrated autonomous football detection and kicking behavior.

Chapter 6: Conclusion

The Humanoid Football-Kicking Robot project successfully demonstrated the integration of embedded systems, computer vision, inverse kinematics, and mechanical design into a functional low-cost humanoid robot. The robot was able to detect a football using a YOLO-based vision system, calculate the required joint angles using inverse kinematics equations, walk toward the target, and perform a kicking motion autonomously after code deployment. The project also showed the effectiveness of using the ESP32 microcontroller, PCA9685 servo driver, and MG995 servo motors in humanoid robotics applications. Although several challenges were encountered during development, including walking instability, drifting, slipping, and motion blur affecting object detection, the system achieved acceptable overall performance and demonstrated successful coordination between software and hardware components. Future improvements may include enhancing balance control, improving gait stability, adding more advanced sensors, and implementing more intelligent motion planning algorithms to increase robot accuracy and reliability.